

The empirical recurrence for OEIS A263120, proved by transfer matrix

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Abstract

OEIS A263120 counts arrays of width 5 over $\{0, 1\}$ whose rows, read as base-2 numbers, are divisible by 15 and whose columns, read the same way, are not divisible by 15. The entry carries an empirical recurrence of order 46 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A column's value depends on every row of the array, so the condition is not local; but its RESIDUE is built one row at a time by Horner's rule, and the vector of column residues is a state of bounded size. The count is then a walk count on $S = 225$ states.

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1 The conjecture

OEIS A263120 is “Number of $(n+3) \times (2+3)$ 0..1 arrays with each row divisible by 15 and column not divisible by 15, read as a binary number with top and left being the most significant bits.”. It has offset 1 and begins

36, 148, 522, 1708, 5124, 15640, 47602, 144236, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 4*a(n-1) - a(n-2) - 8*a(n-3) + 48*a(n-4) - 168*a(n-5) + 48*a(n-6) + 312*a(n-7) - 955*a(n-8) + 2884*a(n-9) - 955*a(n-10) - 4832*a(n-11) + 7632*a(n-12) - 16032*a(n-13) + 7632*a(n-14) + 17568*a(n-15) + 4886*a(n-16) - 72248*a(n-17) + 4886*a(n-18) + 197200*a(n-19) - 366336*a(n-20) + 873744*a(n-21) - 366336*a(n-22) - 1155888*a(n-23) + 1014058*a(n-24) - 588568*a(n-25) + 1014058*a(n-26) - 2290528*a(n-27) + 6020160*a(n-28) - 17209056*a(n-29) + 6020160*a(n-30) + 27546528*a(n-31) - 27604133*a(n-32) + 27776948*a(n-33) - 27604133*a(n-34) + 27085688*a(n-35) - 75797424*a(n-36) + 221932632*a(n-37) - 75797424*a(n-38) - 362608200*a(n-39) + 362785825*a(n-40) - 363318700*a(n-41) + 362785825*a(n-42) - 361187200*a(n-43) + 406335600*a(n-44) - 541780800*a(n-45) + 406335600*a(n-46)$

As of the “Last modified” line on the live entry (Oct 10 2015 07:05:36, revision 4) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 5 cells across and entries $x(i, j) \in \{0, 1\}$, the digits of base 2. Row i of an array with R rows is the number

$$\rho_i = \sum_{j=1}^5 x(i, j) 2^{5-j},$$

the LEFT digit being the most significant, and column j is

$$\gamma_j = \sum_{i=1}^R x(i, j) 2^{R-i},$$

the TOP digit being the most significant. The entry asks that every ρ_i be divisible by 15 and every γ_j be not divisible by 15.

3 Column residues by Horner's rule

The row condition is a condition on one row and is imposed as each row is chosen. The column condition is not local: γ_j depends on every row. What is local is its residue.

Lemma 1. *Let $\gamma_j^{(i)}$ be the value of column j of the first i rows, read the same way. Then $\gamma_j^{(0)} = 0$ and*

$$\gamma_j^{(i)} = 2\gamma_j^{(i-1)} + x(i, j), \quad \text{hence} \quad \gamma_j^{(i)} \bmod 15 = (2(\gamma_j^{(i-1)} \bmod 15) + x(i, j)) \bmod 15.$$

Proof. Adding a row at the BOTTOM shifts every earlier digit one place up in significance, which is multiplication by 2, and puts the new digit in the units place. Reduction modulo 15 commutes with both operations. $\square$

So the vector $(\gamma_1 \bmod 15, \dots, \gamma_5 \bmod 15)$ is a state: at most 15^5 of them, and each array row that satisfies the row condition carries one such vector to another. The walk starts at the all-zero vector, which is the empty array, and the END vector picks out the residue vectors in which every entry is not divisible by 15 — so the end vector is not the all-ones vector, and the column condition is imposed once, at the end, exactly as the entry states it. With M the adjacency matrix,

$$a(n) = \iota^\top M^j \tau, \quad S = 225.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^{46} - \sum_i c_i t^{46-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+46} - \sum_i c_i M^{j+46-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$a(n) = 4a(n-1) - a(n-2) - 8a(n-3) + 48a(n-4) - 168a(n-5) + 48a(n-6) + 312a(n-7) - 955a(n-8) + 2884a(n-9) - 6480a(n-10) + 10240a(n-11) - 12288a(n-12) + 10240a(n-13) - 6480a(n-14) + 2884a(n-15) - 955a(n-16) + 312a(n-17) - 48a(n-18) + 168a(n-19) - 48a(n-20) + a(n-21) - 4a(n-22)$
for every $n > 43$.

Proof. The vector $w = q(M)\tau$ was formed by 46 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 43 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 23 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size, turn each row and each column into a base-2 number with the first digit most significant, and test the divisibility directly. Neither Horner’s rule nor any residue state is involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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